

Architecture & Security Constraints

Project Boomerang (Self-Service Returns) | Author: Lena Fischer, Enterprise Architecture | Issued 21 March, in response to BRD v0.9

1. Purpose

This note records the constraints the returns portal must be designed within. Several of them are incompatible with the service levels stated in the Business Requirements Document; those are called out explicitly in section 5 and need a decision from the steering group before build starts.

2. Systems of record

- SAP ECC remains the system of record for stock and for the financial posting of a credit note. The portal must not write to SAP in phase 1.
- Stock and goods-receipt data reach downstream systems through a nightly IDoc feed that completes at 23:30 CET. There is no real-time interface and none is planned before 2027.
- Adyen is the payment service provider for all consumer channels. Refunds are initiated through the Adyen API and settle on a **T+7** basis; the amount is not visible on the customer's statement until settlement completes.
- Carrier label generation for DHL and GLS runs as a **batch job at 02:00 CET**. Neither carrier contract includes access to a synchronous label API.

3. Platform and security

- Customer authentication must use the existing Azure AD B2C tenant. No new identity store may be introduced.
- All data must be stored and processed in the EU, in the Frankfurt region.
- PostgreSQL is the only approved relational database. No new database technology will be provisioned for this project.
- Any new public endpoint must pass the Security Review Board, which requires a 10 working day lead time and a completed threat model.
- All customer-facing traffic must route through the existing web application firewall.

4. Non-functional constraints

Area	Constraint
Throughput	Must sustain 19,000 return requests per month with a peak of 400 concurrent users, without horizontal scaling beyond four nodes.
Audit	Every automated accept, reject or fee decision must be reproducible on demand for 7 years . This is a financial-audit obligation and is independent of the marketing-consent lifecycle.
Observability	All services must emit traces to the existing OpenTelemetry collector.
Release	Deployments are blocked between 15 December and 6 January, the peak trading freeze.

5. Conflicts with the Business Requirements Document

The following BRD statements cannot be met as written. Each needs an explicit decision.

BRD ref	Stated requirement	Constraint	Consequence
BR-04	Return label generated immediately	Carrier label batch runs at 02:00 CET	A label requested at 09:00 is not available until the following morning unless the carrier contract is renegotiated.
BR-06	Refund issued within 5 business days	Adyen settles T+7	The refund can be <i>initiated</i> within 5 business days but cannot be <i>received</i> by the customer within that window. The requirement conflates two different events.
Sec. 6	Personal data retained 24 months	Audit records retained 7 years	The two retention periods apply to overlapping data. A position on pseudonymisation is required from Legal.

6. Assumptions

- Phase 1 is delivered by four engineers and one designer over one quarter.
- The portal is a new service and does not modify the existing order-history application beyond adding an entry point.
- Marketplace orders are held in a separate schema and are not addressed by this note.